

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0133

Fielder Creek 100 kV
November 29, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0133. The Interconnection Customer proposes a 114.9 MW wind facility interconnecting at the Fielder Creek 100 kV substation in the Illinois area. The request was received on June 04, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 95 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,340,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Deltrail / Fielder Creek 100 KV line
Contingency: Fielder Crk Substation / Curry Rd 100 KV
Loading: 109.8% (Rating: 131 MVA)
DFAX: 64.2%
Pre-project loading: 53.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deltrail / Fielder Crk 100 KV	Reconductor line	\$2,284,000	59% (\$1,340,000)
	Total allocated to project		\$1,340,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.